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Q1. Write a program that uses a while loop to calculate

Q2. Write a program that uses a while loop to find the

- `num = 6`

Q3. Write a program using a while loop that prints num

Q4. Write a while loop to print the first 20 even number

Q5. Write a program that uses a while loop to find the

- `num = 54321`

Q6. Write a program that reverses a number using a w

Q7. Create an infinite while loop that asks the user to e
breaks; otherwise, it prints the square of the number?

Q8. write a loop statement to print the following series

- output : 10,20,30,40,....300
- append them to a list

Q9. write a while loop statement to print the following

- output : 105,98,91.....7.
- append them to list.

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Q16. create an empty list. Continuously ask the user to until they type "done." Use the append() method to add

Q17. write a program to accept 10 numbers from the us

Q18. write a program to print all the characters in strin

Q19. write a python program to print only the odd num

- `list = [23,45,25,32,46,33,71,90]`.

Q20. write a python program to get the following outp

- 1--- 49
- 2--- 48
-
- 49--- 1

Q21. Write a program that uses a while loop to check

- `str = "madam"`

Q22. Write a program using a while loop to print all th

Q23. Write a program to find the least common multip

Q24. Write a program using a while loop to print a rigl

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Q1. Write a Python function to find the sum of all elements in a list.

Q2. Write a Python program to count the occurrence of each element in a list.

Q3. Write a Python code to generate a list of squares for numbers from 1 to 10.

Q4. Write a Python function that:

- Takes two lists as input.
- Merges the two lists into a single list, alternating elements from each list.
- Returns the merged list.

Q5. Given a list of numbers `numbers = [1, 2, 3, 4, 5, 6, 7]`, write a function to remove multiple elements from a list?

- to remove 3, 5 and 7

Q6. Given a list of integers `[10, 20, 30, 20, 20, 30, 40]`, write a function to remove duplicates. Generate another list, which contains only the unique elements.

Q7. Given 3 digits 4, 5, and 6. Find all the possible combinations of these digits.

Q8. Write a Python program to create a set of even numbers from 1 to 100.

Q9. Write a Python program to find the union of two sets using the `union()` method?

Q10. Given a list of integers, write a function that partitions the list into two parts: one for odd numbers and one for even numbers.

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Q18. covert the following loop into for loop :

- `x = 4`
- `while (x<=8)`
- `print(x*10)`
- `x+=2`

Q19.Python program to check the validity of password

Q20. Create a program that prints each element of a li

Q21. Write a program that prints all numbers between

Q22. Write a program that accepts a string from user a
letters using a for loop?

Q23.Write a program that extracts all digits from a giv

Q24. Given a list of names, write a program that creat
lengths are values using a for loop?

- `names = ["Alice", "Bob", "Charlie", "David", "`

Q25.Accept two strings from user, write a program th
duplicates) using a for loop?

Q26. Write a program that generates a list of n random

Q27.Write a Python program to reverse the order of cl
built function?



Q1. write a program if a list contains a palindrome of elements or not?

- `l = [1,2,3,2,1]`

Q2. write a program to check whether 13 is a prime number or not?

Q3. Python program to convert the month name to a number of days:

Q4. Take value of length and breath from the user and check if it is square or not?

Q5. Take two numerical values from user and print greatest among them?

Q6. Calculate the percentage of class attend and say who are eligible and who are not eligible to attend exam?

- `working_days = 280`

Q7. write a program to accept percentage from the user and display category accordingly?

- `< 40 -- Failed`
- `<55 & >=40 -- Fair`
- `<65 & >=55 -- Good`
- `greater or =65 -- Excellent`

Q8. Accept three sides of a triangle from user and check whether the triangle is equilateral, isosceles or scalene triangle?

- note :
- equilateral triangle has three equal sides
- isosceles triangle has at least two equal sides
- scalene triangle has three unequal sides

Q9. Accept the marks from the user of the subjects English, Math, Science, Social studies and display the stream allotted?

- All subjects more than 80 marks -- Science Stream
- English > 80 and Math, Science above 50 -- Commerce Stream
- English > 80 and Social Studies > 80 -- Humanities

Q10. Write a function that takes a student's score as input and returns the grade based on the following criteria:

- A: 90-100
- B: 80-89
- C: 70-79
- D: 60-69
- F: Below 60



Q11. Write a program that takes employees salary as an input to calculate the bonus of an employee based on their years of service . The program should work as follows:

- If the years of experience is less than 5 : print(" no bonus credited")
- if the years of experience is more than 5 : calculate the bonus as 5 % of salary and diplay the amount credited.

Q12. write a program to accept two numbers from the user and a mathematical operator and perform function accordingly?

Q13. write a program to accept the three sides of a triangle and check whether it is a triangle or not?

Q14. create a program that calculates the library charges based on the number of days a book is borrowed and not returned. The charge structure is as follows:

- For exactly 5 days: Rs 2 per day
- For 6 to 10 days: Rs 3 per day
- For 11 to 15 days: Rs 4 per day
- For more than 15 days: Rs 5 per day

Q15. Write a program to check whether a person is eligible for voting or not and also eligible for getting driving license or not?

- criteria for voting : more than or equal to 18 - eligible to vote
- criteria for getting driving license: more than or equal to 21

Q16. write a program to check whether the number accepted from user is divisible by 7 or not?

Q17. write a program to print "Hello" if the number entered by user is multiple of 5 otherwise print "Bye!"?

Q18. write a program to display the last digit of the number entered by the user?

Q19. write a program to display the last digit of the number entered by the user and check whether it is divisible by 3 or not?

Q20. write a program to check whether year is leap year or not ?

Q21. write a program to check whether a number entered by user is divisible by both 2 and 3?

Q22. Accept the age of four people and display the youngest in them?

Q23 write a program to accept the age, gender, no of days of work from the user and display wages accordingly?

- gender : male & age ≥ 18 and < 30 wage_perday = 700
- gender : female & age ≥ 18 and < 30 wage_perday = 750
- gender : male & age ≥ 30 and < 40 wage_perday = 800
- gender : female & age ≥ 30 and < 40 wage_perday = 850

Q24. write a program if a list contains a palindrome of elements or not?



- l= [1, "abc",1,"abc"]

Q25. Write a program that prompts the user to enter their age and then outputs

whether they are:

- A child (0-12)
- A teenager (13-19)
- An adult (20-64)
- A senior (65 and above)

Q26. Write a program that asks the user to enter a day of the week (as a string) and determines if it is a weekend (Saturday or Sunday) or a weekday?

Q27. Write a function that calculates the factorial of a given number using recursion?

Q28. Create a program that takes a number and prints whether it is positive, negative, or zero?

Q29. create a program to accept the kilometers covered and calculate the bill according to following criteria :

- First 10 km --- rs 11/km
- Next 90 km --- rs 10/km
- After that --- rs 9/km

Q30. Accept the market price from the user and calculate the net amount to be paid?

- amount to be paid = market_p - (market_p * discount)
- market_p > 10,000 --- discount = 20%
- market_p > 7000 & < 10,000 --- discount = 15%
- market_p <= 7000 --- discount = 10%

